

Stage 1 — Project scaffold

initial project setup Create the repo structure: `src/`, `data/`, `notebooks/`, `outputs/`, `requirements.txt`, `.gitignore`, and a bare-bones `README.md` describing the project goal.

add manga109 data loader Write a script to load Manga109 annotations (their XML format) and map them to image paths. Just loading and printing basic stats at this point.

add basic visualization utility A helper function to draw bounding box annotations on a page image using matplotlib. Useful for sanity-checking labels throughout development.

Stage 2 — Preprocessing

explore binarization approaches in notebook A Jupyter notebook trying Otsu thresholding, adaptive thresholding, and simple global thresholding on a few sample pages. No conclusions yet — just exploration.

implement binarization and noise reduction pipeline Settle on adaptive thresholding + morphological opening for noise. Write it as a reusable `preprocess.py` module.

fix binarization failing on light-background pages A targeted bugfix — some Manga109 pages have near-white backgrounds that the thresholding was inverting incorrectly.

Stage 3 — Panel detection (edge-based)

add canny edge detection for panel boundaries Apply Canny to preprocessed images and visualize the raw edge maps. Lots of noise and fragmented lines at this stage.

add morphological closing to repair broken panel edges Dilate + close the edge map to connect fragmented border segments into more continuous loops.

add contour extraction and geometric filtering Extract contours from the closed edge map, filter by area and aspect ratio, discard obvious non-panels.

add polygon approximation for irregular panel shapes Use `approxPolyDP` to fit polygons to contours instead of bounding boxes, since manga panels are rarely rectangular.

evaluate panel detection on manga109 validation split Compute IoU between predicted panel regions and ground-truth annotations. First real numbers — probably not great, which motivates next steps.

improve contour filtering: add closure score heuristic Panels with broken borders were slipping through. Add a contour closure check (ratio of arc length to convex hull perimeter) to filter them better.

Stage 4 — YOLO fine-tuning for speech bubbles

convert manga109 annotations to yolo format Script to convert Manga109 XML annotations into YOLO `.txt` label files (normalized xywh per line). Separate train/val/test splits.

add yolo training config and dataset yaml Write the `dataset.yaml` and initial `train.py` script using Ultralytics YOLOv8. First training run at this point.

experiment: compare yolov8n vs yolov8s for bubble detection A short notebook comparing nano vs small model size — note tradeoff between speed and mAP on the val set.

tune yolo hyperparameters: increase mosaic augmentation Manga pages are dense and high-contrast; mosaic augmentation helps. Log the mAP improvement.

add inference script for speech bubble detection Clean `detect_bubbles.py` that runs the fine-tuned YOLO model on a page and returns bounding boxes.

Stage 5 — Integration and output

add panel-to-bubble assignment using iou overlap For each detected bubble, assign it to the panel with highest IoU overlap. Handle edge cases where a bubble overlaps multiple panels.

add speech bubble tail direction heuristic Detect the approximate direction of the bubble tail (pointing toward a character) using contour convexity defects. Used to improve cross-panel attribution.

build hierarchical page representation Combine panel detections and bubble assignments into a structured `Page` object: panels as nodes, bubbles as children, adjacency as edges.

add reading order inference from panel adjacency graph Use a simple top-left to bottom-right traversal of the panel adjacency graph to infer reading order.

add end-to-end pipeline script `run_pipeline.py` that takes an image path and outputs the full structured representation + a visualization.

Stage 6 — Evaluation and cleanup

add full evaluation script with precision recall and iou Compute per-page and aggregate metrics for both panel detection and bubble detection against Manga109 ground truth.

run evaluation on test set, log results Actual numbers. Record them somewhere (a **results/** folder or wandb) — you'll need these for the paper.

add qualitative failure case analysis notebook A notebook stepping through pages where the pipeline fails: diagonal panels, very dense pages, irregular bubble shapes. Good material for the limitations section.

refactor: clean up src structure and add docstrings Cleanup commit before writing — consolidate modules, add function-level documentation.

update readme with setup instructions and results summary Final README with how to run everything, dataset setup steps, and a table of your key results.